

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.

4. Determine the remaining unused IP address range within the given network block.

SOL:: 1. School of Computing (/25)

- Subnet Mask: 255.255.255.128
- Network Address: 192.168.100.0
- Broadcast Address: 192.168.100.127
- Valid Host Range: 192.168.100.1 – 192.168.100.126
- Total Addresses: 128
- Usable Host Addresses: 126

2. School of Electronics (/26)

The next available block starts at 192.168.100.128.

- Subnet Mask: 255.255.255.192
- Network Address: 192.168.100.128
- Broadcast Address: 192.168.100.191
- Valid Host Range: 192.168.100.129 – 192.168.100.190
- Total Addresses: 64
- Usable Host Addresses: 62

3. School of Mechanical Engineering (/27)

The next available block starts at 192.168.100.192.

- Subnet Mask: 255.255.255.224
- Network Address: 192.168.100.192
- Broadcast Address: 192.168.100.223
- Valid Host Range: 192.168.100.193 – 192.168.100.222
- Total Addresses: 32
- Usable Host Addresses: 30

4. Remaining Unused IP Address Range

After allocating the three subnets:

- /25 → 192.168.100.0 – 192.168.100.127
- /26 → 192.168.100.128 – 192.168.100.191
- /27 → 192.168.100.192 – 192.168.100.223

The remaining unused addresses are:

- Unused Network Range: 192.168.100.224 – 192.168.100.255
- Total Remaining Addresses: 32

This remaining block is equivalent to a /27 subnet.

FINAL ANSWERS

Department	Subnet Mask	Network Address	Host Range	Broadcast Address	Usable Hosts
School of Computing	of/25 (255.255.255.128)	192.168.100.0	192.168.100.1 – 192.168.100.126		126
School of Electronics	of/26 (255.255.255.192)	192.168.100.128	192.168.100.129 – 192.168.100.190		62
School of Mechanical Engineering	of/27 (255.255.255.224)	192.168.100.192	192.168.100.193 – 192.168.100.222		30

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

SOL:

Department	Subnet Mask	Network Address	Valid Range	HostBroadcast Address	Usable Hosts
Software Development	/24 (255.255.255.0)	10.10.0.0	10.10.0.1 10.10.0.254	–10.10.0.255	254
Quality Assurance	/25 (255.255.255.128)	10.10.1.0	10.10.1.1 10.10.1.126	–10.10.1.127	126
Human Resources	/26 (255.255.255.192)	10.10.1.128	10.10.1.129 10.10.1.190	–10.10.1.191	62
Executive Management	/27 (255.255.255.224)	10.10.1.192	10.10.1.193 10.10.1.222	–10.10.1.223	30

Unused IP Address Range

After allocating all four subnets, the allocated addresses end at:

- Last allocated IP: 10.10.1.223

The remaining unused addresses are:

Description	IP Address Range
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FINAL ANSWER:

Unused IP Range 10.10.1.224 – 10.10.255.255

Department	Subnet Mask	Network Address	Host Range	Broadcast Address	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.1 10.10.0.254	– 10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1 10.10.1.126	– 10.10.1.127	126
Human Resources	/26	10.10.1.128	10.10.1.129 10.10.1.190	– 10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193 10.10.1.222	– 10.10.1.223	30
Unused Range	IP —	10.10.1.224	10.10.1.224 10.10.255.255	– 10.10.255.255	Remaining addresses 10.10.0.0/16 block

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

SOL: Given IPv6 Network Address

2001:0db8:abcd:0000:0000:0000:0000/64

1. Convert the given IPv6 address into its compressed notation

Rules:

- Remove leading zeros in each hexadecimal block.
- Replace consecutive groups of 0000 with :: (only once).

Compressed IPv6 Address

2001:db8:abcd::/64

2. Expand the compressed address back into its full hexadecimal representation

Compressed Address:

2001:db8:abcd::/64

Expanding :: back to all eight 16-bit blocks:

2001:0db8:abcd:0000:0000:0000:0000/64

3. Identify the network prefix using the /64 prefix length

A /64 prefix means:

- First 64 bits = Network Prefix
- Last 64 bits = Interface Identifier (Host Portion)

Network Prefix

2001:0db8:abcd:0000::/64

or simply

2001:db8:abcd:0::/64

4. Calculate the total number of host addresses supported within the subnet

Host bits = $128 - 64 = 64$ bits

Total host addresses:

2^{64}

= 18,446,744,073,709,551,616

Total Host Addresses

18,446,744,073,709,551,616 addresses

5. Total number of possible host addresses available within this network

Since this is a /64 IPv6 subnet:

$$2^{64}$$

$$= 18,446,744,073,709,551,616$$

Answer: 18,446,744,073,709,551,616 possible host addresses.

Final Answer:

Task	Answer
Given IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000/64
Compressed Notation	2001:db8:abcd::/64
Expanded Notation	2001:0db8:abcd:0000:0000:0000:0000/64
Network Prefix (/64)	2001:db8:abcd:0::/64
Host Portion	Last 64 bits
Total Host Addresses	$2^{64} = 18,446,744,073,709,551,616$
Total Possible Host Addresses	18,446,744,073,709,551,616

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

SOL:

Given Routing Table

Network Address Prefix

192.168.1.0 /24

192.168.2.0 /24

192.168.3.0 /24

Destination IP Address

192.168.2.45

1. Determine the destination subnet to which the packet belongs

For a /24 subnet mask (255.255.255.0), the first three octets represent the network portion.

Destination IP:

192.168.2.45

Network Address:

192.168.2.0/24

Answer: 192.168.2.0/24

2. Identify the matching routing table entry

Compare the destination network with the routing table:

Routing Table Entry Match

192.168.1.0/24 ✗ No

192.168.2.0/24 ✓ Yes

192.168.3.0/24 ✗ No

Matching Routing Table Entry:

192.168.2.0/24

3. Determine the next-hop network/interface used to forward the packet
4. Default route used if no routing table entry matches

If the destination IP does not match any routing table entry, the router uses the default route.

The default route is represented as:

0.0.0.0/0

or

0.0.0.0 0.0.0.0

The packet is forwarded to the default gateway (next-hop router) configured by the network administrator.

Conclusion

Task	Answer
Destination IP Address	192.168.2.45
Destination Subnet	192.168.2.0/24
Matching Routing Table Entry	192.168.2.0/24
Next-Hop Network/Interface	Interface connected to 192.168.2.0/24 (e.g., G0/1)
Default Route (if no match exists)	0.0.0.0/0 (Default Gateway)

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

SOL:

Given

- Administration LAN: 192.168.10.0/26
- Finance LAN: 192.168.10.64/26

Router Interfaces (Gateways):

- Administration Gateway → 192.168.10.1
- Finance Gateway → 192.168.10.65

A /26 subnet mask is 255.255.255.192, which provides:

- 64 IP addresses per subnet
- 62 usable host addresses

1. Determine the Network Address and Broadcast Address

LAN	Network Address	Broadcast Address
Administration	192.168.10.0/26	192.168.10.63
Finance	192.168.10.64/26	192.168.10.127

3. Suggest Suitable IP Addresses for Three Computers in Each LAN

Administration LAN

Device	IP Address	Default Gateway
Computer 1	192.168.10.2	192.168.10.1
Computer 2	192.168.10.3	192.168.10.1
Computer 3	192.168.10.4	192.168.10.1

Finance LAN

Device	IP Address	Default Gateway
Computer 1	192.168.10.66	192.168.10.65
Computer 2	192.168.10.67	192.168.10.65
Computer 3	192.168.10.68	192.168.10.65

CONCLUSION:

The network has been successfully divided into two separate /26 subnets to support the Administration and Finance departments. Each subnet contains 64 IP addresses, of which 62 are available for host devices, while one address is reserved as the network address and another as the broadcast address.

The Administration LAN uses the subnet 192.168.10.0/26, where the network address is 192.168.10.0, the broadcast address is 192.168.10.63, and the valid host addresses range from 192.168.10.1 to 192.168.10.62. The router interface 192.168.10.1 acts as the default gateway, and the computers can be assigned addresses such as 192.168.10.2, 192.168.10.3, and 192.168.10.4.

Similarly, the Finance LAN uses the subnet 192.168.10.64/26, where the network address is 192.168.10.64, the broadcast address is 192.168.10.127, and the valid host addresses range from 192.168.10.65 to 192.168.10.126. The router interface 192.168.10.65 serves as the default gateway, and the computers can be assigned addresses such as 192.168.10.66, 192.168.10.67, and 192.168.10.68.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts	Applies concepts	Limited or partially	Unable to apply

		effectively to address the scenario	with minor errors	correct application	relevant concepts
Decision Making	20%	Makes well-reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand